

Machine Learning Regression method using R^2 value :

1. Multiple Linear Regression (R^2 Value) = 0.87526
2. Support Vector Machine:
With Standard Scaler R Score :

S.No	Hyper Parameter	Linear (r value)	RBF (Non Linear) (r value)	POLY (r value)	SIGMOID (r value)
1	C=0.1, gamma=auto	-0.0573067	-0.0574795	-0.0574480	-0.057458
2	C=10, gamma=auto	-0.0396449	-0.05680	-0.05366	-0.05471
3	C=100, gamma=scale	0.106468	-0.050726	-0.01980	-0.03045
4	C=1000, gamma=0.1, epsilon=0.1	0.780283	0.02927	-0.0105603	0.094362
5	C=1000	0.78028	0.00676	0.26616	0.1850686
6	C=1000, gamma=auto, degree=3, epsilon=0.1, coef0=1	0.780283	0.00676	0.682406	0.10497

The SVM Regression use R^2 Value

3. Decision Tree:

criterion	splitter	max_depth	min_samples_split	min_samples_leaf	r2_score
absolute_error	best	20	2	2	0.956486
absolute_error	best	NaN	2	2	0.956486
absolute_error	best	5	2	2	0.956486
absolute_error	best	10	2	2	0.956486
absolute_error	best	20	5	1	0.950717
absolute_error	best	NaN	5	1	0.950717
absolute_error	best	5	5	1	0.950717
absolute_error	best	10	5	1	0.950717
absolute_error	best	NaN	2	1	0.946457
absolute_error	best	20	2	1	0.946457

Decision Tree Regression use R^2 Value : 0.956486

4. Random Forest :

n_estimators	max_depth	min_samples_split	min_samples_leaf	max_features	bootstrap	r2_score
50	5	2	1	None	TRUE	0.910615
200	5	2	1	None	TRUE	0.907634
50	10	2	1	None	TRUE	0.907536
50	NaN	2	1	None	TRUE	0.907536
50	20	2	1	None	TRUE	0.907536
200	10	2	1	None	TRUE	0.907451
200	20	2	1	None	TRUE	0.907451
200	NaN	2	1	None	TRUE	0.907451
100	5	2	1	None	TRUE	0.904393
100	20	2	1	None	TRUE	0.904192